

Iterative process: We follow same instructions in several rounds.

Solve opt prob. $\Rightarrow$ max/min $f_n = f(\dots)$

view the soln as a seq $(x_i, x_j, \dots)$
 $(x_1, x_2, \dots, x_n)$

We ~~not~~ choose one of the many possible options "Greedy".

Our choice would be such that contributes more in the obj fn. if it is a max prob.

The Scheduling Problem

Input: A set of n jobs $\{J_1, J_2, \dots, J_n\}$ with positive lengths $\{l_1, l_2, \dots, l_n\}$ and positive wts $\{w_1, w_2, \dots, w_n\}$.

Output: A job seq that minimizes the sum of wted completion time.



$$\begin{aligned} J &= \overbrace{(J_2, J_{10}, J_5, \dots, J_9)}^{l_2 + l_{10} + l_5} \\ J' &= (J_5, \dots, J_{10}) \end{aligned}$$

Completion time of a job (j) in a schedule δ .

$C_j(\delta)$ = This is the sum of the lengths of the jobs preceding j in δ , plus the length of j itself.

Weighted completion time of (j) in δ :

$$= w_j \times C_j(\delta)$$

The sum of weighted completion time in a schedule δ .

$$C(\delta) = \sum_{j=1}^n w_j \cdot C_j(\delta)$$

Example:

	J_1	J_2	J_3
length	1	2	3
wt's	3	2	1

$$\delta_1 = \begin{pmatrix} 1.3 + 2.5 + 3.6 \\ J_3, J_2, J_1 \\ \textcircled{3} \textcircled{5} \textcircled{6} \end{pmatrix}$$

$$C(\delta_1) = \underline{\underline{31}}$$

$$\delta_2 = \begin{pmatrix} 2 & 1 & 3 \\ J_2 & J_3 & J_1 \\ \textcircled{2} \textcircled{5} \textcircled{6} \end{pmatrix}$$

$$C(\delta_2) = 4 + 5 + 18 = \underline{\underline{27}}$$

$$\delta_3 = \begin{pmatrix} 3 & 2 & 1 \\ J_1 & J_2 & J_3 \\ \textcircled{1} \textcircled{3} \textcircled{6} \end{pmatrix}$$

$$C(\delta_3) = 3 + 6 + 6 = \underline{\underline{15}}$$

Case I All job weights are identical. (say w)

$$\mathcal{J} = (J_1, J_2, \dots, J_n)$$

$$\begin{aligned} C(\mathcal{J}) &= w \cdot l_1 + w(l_1 + l_2) + w(l_1 + l_2 + l_3) + \dots + w(l_1 + l_2 + \dots + l_n) \\ &= w \left[n \cdot l_1 + (n-1)l_2 + (n-2)l_3 + \dots + 1 \cdot l_n \right] \end{aligned}$$

$\Rightarrow$ We should schedule shorter length jobs earlier.

Case II: All job lengths are identical. (say l)

$$\mathcal{J} = \begin{pmatrix} J_i & J_k & J_p & \dots & J_t \\ l & 2l & 3l & & nl \end{pmatrix}$$

$$C(\mathcal{J}) = l(w_i \cdot 1 + w_k \cdot 2 + w_p \cdot 3 + \dots + w_t \cdot n)$$

$\Rightarrow$ We should schedule larger wt jobs earlier.

For general case, we assign a "score" to each jobs so that scheduling the job from highest to lowest score is guaranteed to minimize the sum of the wted completion time.

Jobs	J_1	J_2	$\dots$	J_n
Score	s_1	s_2		s_n

(decr. order)

If $(J_1, J_2, \dots, J_n)$ is the schedule obtained by the decreasing ordering of the scores, then that ordering should also satisfy the two special cases, that is, decreasing order of w_j (in case II) and also increasing order of lengths (in case I).

Score:

Greedy diff

Schedule the jobs with decreasing order of $\underline{s_j = \frac{w_j \cdot l_j}{l_j}}$

Case I: $\left(\begin{array}{l} \text{Suppose } w_j = w \ \forall j \\ s_j = w - l_j \\ \text{dec order of } s_j \Leftrightarrow \text{inc order of } l_j \end{array} \right)$

Case II: $\left(\begin{array}{l} \text{Suppose } l_j = l \ \forall j \\ s_j = w_j - l \\ \text{dec ord of } s_j \Leftrightarrow \text{dec order of } w_j \end{array} \right)$

Greedy Ratio

Schedule the jobs in dec order of $s_j = \frac{w_j}{l_j}$

Example:

	J_1	J_2
length	5	2
wt	3	1

$$\underline{G.\text{diff}} = w_j - l_j \quad \begin{array}{|c|c|} \hline -2 & -1 \\ \hline \end{array}$$

$$G.\text{ratio} = w_j / l_j \quad \begin{array}{|c|c|} \hline \frac{3}{5} = .6 & \frac{1}{2} = .5 \\ \hline \end{array}$$

$$\mathcal{J}_{G.\text{diff}} = \begin{pmatrix} J_2 & J_1 \\ \textcircled{2} & \textcircled{7} \end{pmatrix}$$

$$C(\mathcal{J}_{G.\text{diff}}) = 2 + 21 = 23$$

$$\mathcal{J}_{G.\text{Ratio}} = \begin{pmatrix} J_1 & J_2 \\ \textcircled{5} & \textcircled{7} \end{pmatrix}$$

$$C(\mathcal{J}_{G.\text{Ratio}}) = 15 + 7 = 22$$

Q: Can we argue from this example that Greedy ratio will give the min sum of wted completion time??

Ans: NO, We can only argue that Greedy ~~ratio~~ diff fails.

(*) Greedy Ratio always gives the correct schedule.

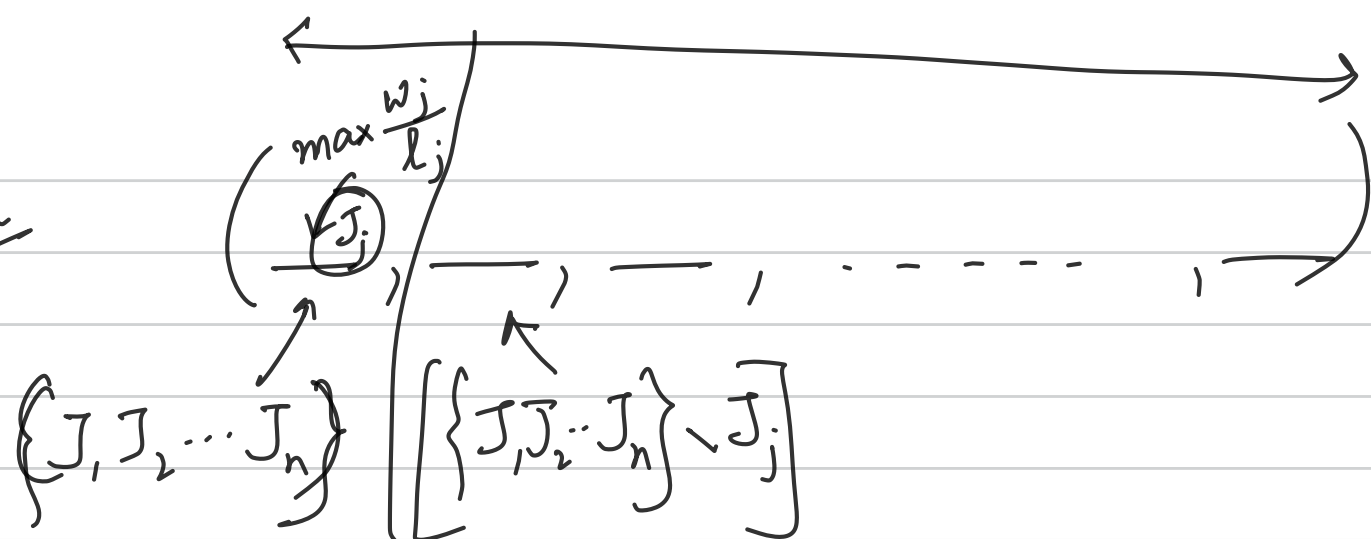
We need to prove this fact.

Algo

- $O(n)$ ① Assign score $s_j = w_j / l_j$ for every job j .
- $O(n \log n)$ ② Sort the jobs according to the dec. order of scores.
- ③ Return $(\mathcal{J})$.

Running time: $O(n \log n)$

Soln



$$\frac{wt}{length} \Rightarrow \text{wt per unit time.}$$

Scheduling the jobs according to the dec order of $wt/length$ means that we are putting the job that has max wt per unit time first.

Obs. the processor works for $\sum_{i=1}^n l_i$ time and this is independent of σ .

